

Is This X ? A Null-Standardized Question Space for Corpus Curvature, with a Hodge-Theoretic Channel and a Calibrated Standard-Candle Generator

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Abstract

Thirty-four measurement cycles of a corpus-curvature program asked “is this X ?” one X at a time — Ginzburg-Landau, LIGO, Grothendieck, pentagon — and answered PARTIAL every time, because no cycle computed what its statistic would have been under a matched null. This paper replaces the sequence of candidate X ’s with a map. Four corrections and three constructions are reported. (i) The headline observable $V_2 = PC1 + PC2$ is a decreasing function of feature-space dimension D at fixed rows, so the celebrated $+0.4664$ “phase transition” ($0.2993 \rightarrow 0.7657$) is a basis change: on the identical 2286 rows we measure $V_2 = 0.7235$ at $D = 9$ and 0.2940 at $D = 24$. (ii) the shipped estimator reports not the participation ratio but the two-share proxy $\hat{r} = 2/V_2$, so the gate $V_2 \geq 0.40$ is by definition $\hat{r} \leq 5$ — an identity with no empirical content; what is empirical is that the exact-1.0000 gate passes at 2-D and at the 384-D ($\ell = 384, m = 3$) variant are rank degeneracies reproduced with zero data content on all six corpora, including a single-class iid one. (iii) Under a fixed-margin (curveball) null drawn by two independent, provably uniform samplers (χ^2 uniformity $p = 0.364$ and $p = 0.506$ on exhaustively enumerated fibres), the 2286×9 matrix has null $V_2 = 0.7089 \pm 0.0014$ against a real 0.723529 : $z = +12.3$. About 98% of V_2 is fixed by the marginals and the surviving signal is $\Delta V_2 = +0.0144$. An intermediate analysis reported 0.8005 , $z = -45.8$; that value is retracted here after two independent replications. (iv) $\Delta V_2(N)$ is flat ($+0.0135$ to $+0.0175$) for $N \geq 79$ with no interior maximum, so the previous $N_c \approx 40$ was the null’s own finite-size inflation and the transition does not exist. We add a Hodge channel (Jiang-Lim-Yao-Ye decomposition on the item graph): $\rho_g = 0.99961$, $\rho_c = 0.00013$, $\rho_h = 0.00026$, $\beta_1 = 6756$, and against a degree-matched null $z(\rho_h) = -53.1$, $z(\beta_1) = -312.5$ — the corpus is strongly *anti*-cyclic, which inverts the vortex reading; all three pre-registered Hodge predictions are falsified. The positive result is an instrument: a standard-candle generator planting a spherical-harmonic curve of amplitude s on a Fibonacci golden-angle lattice, on which the null-standardized statistic $\Delta V_2 z$ has detection power monotone in s over $s \geq 0.5$ at every grid point and a bounded false-positive rate ($s = 0$: $z \approx 0$, one-sided FPR 8.3% against a nominal 5%; $s = 1$, $N = 512$: $z = +3.43$, power 1.00; $s = 2$, $N = 2048$: $z = +28.07$), while at $s \leq 0.25$, and at $N = 128$ throughout, the cell means sit at the power floor and are not ordered while raw V_2 misorders corpora by signal. The deliverable is *IS-THIS-X*: a map Φ from corpus to null-standardized statistic vector, reference families, exclusion-only verdicts. yubiOS places as M_0 -excluded ($z = +12.3$), nearest a two-population mixture with a small positive residual; mean-field Ginzburg-Landau is excluded.

Introduction

The predecessor of this paper (Tchatalbachian 2026a) fitted learned latent curves to a binary corpus-coverage matrix and reported one fit-quality gate: the top-2 eigenvalue share $V_2 = PC1 + PC2 \geq 0.40$. Thirty-four subsequent measurement cycles took that number as an order parameter and asked, in sequence, whether the corpus *is* a Ginzburg-Landau system (Tchatalbachian 2026b; Hohenberg & Krekhov 2015), a gravitational-wave catalogue, a Grothendieck-gap instance, a pentagon. Each cycle answered by narrative compatibility. The median verdict was PARTIAL.

The defect is structural. A question space with candidate classes but no origin cannot weigh “compatible with X ” against “compatible with nothing”. Every class was tested with bespoke estimators — saturation windows, kernel bandwidths, row-lag correlation lengths — not comparable across X , across corpora, or even across cycles; the same κ estimator returned 1.06 on one corpus and 323 on another, the latter a saturation-clamp artifact. The fix is therefore not another X . It is a map,

$$\Phi: \text{corpus} \mapsto s \in \mathbb{R}^m, \quad s \text{ null-standardized}, \quad (1)$$

with the matched-marginal null at the origin, invariance requirements that delete non-invariant estimators by construction, and exclusion-only verdicts. Equation (1) is the thesis; Sections 3–6 are its content.

Section 2 restates the predecessor’s governing equations, with the Y_3^3 normalization corrected. Section 3 gives the null models and the corrected record, including a retraction. Section 4 gives the Hodge channel and its three falsified predictions. Section 5 gives the standard-candle generator and calibrated detection — the paper’s positive result. Section 6 defines the *IS-THIS-X* question space and places yubiOS and a GWTC-like corpus in it. Sections 7–8 are scope and conclusion; Appendix A is the reproduction manifest.

Two disciplines run throughout. No statistic is reported without the value it takes on a matched null. And when a number in the internal record fails to reproduce, the failure is printed rather than the number quietly replaced — Section 3.3 is that discipline applied to this program’s own most-cited intermediate result.

Background: the learned-latent-curve family

Flat curve, ridge solve, gate

For output dimension $j = 1, \dots, D$ and a 1-D ordering coordinate $t \in [0, 1]$ (Tchatalbachian 2026a),

$$z_j(t) = a_{j,0} + \sum_{m=1}^k (a_{j,m} \sin(2\pi f_m t) + b_{j,m} \cos(2\pi f_m t)), \quad (2)$$

with k shared learned frequencies. Stacking the design vector

$$\phi(t) = [1, \sin(2\pi f_1 t), \cos(2\pi f_1 t), \dots, \sin(2\pi f_k t), \cos(2\pi f_k t)] \in \mathbb{R}^{1+2k} \quad (3)$$

gives $y(t) = C\phi(t)$, solved at fixed frequencies by the Tikhonov ridge

$$C^* = \left(\Phi^\top \Phi + \lambda I \right)^{-1} \Phi^\top Z, \quad \lambda = 10^{-3}, \quad (4)$$

still the sanity floor for any gradient fit at the same frequencies. The coordinate t is the rank of the top principal component of the z -scored feature matrix, and the gate is

$$V_2 = PC1 + PC2 = \frac{\lambda_1 + \lambda_2}{\sum_{a=1}^D \lambda_a} \geq 0.40, \quad (5)$$

the λ_a being eigenvalues of the $D \times D$ correlation matrix, so $\sum_a \lambda_a = D$. Equation (5) is the observable this paper audits.

Hyperspherical variant and Möbius chart

For $\mathbf{x} \in S^2$,

$$y(\mathbf{x}) = \mathbf{b} + \sum_{\ell=0}^L \sum_m \mathbf{a}_{\ell,m} Y_{\ell,m}(\varphi_\theta(\mathbf{x})), \quad n_{basis} = \sum_{\ell=0}^L (2\ell + 1) = 16 \text{ at } L = 3, \quad (6)$$

with the Möbius reparameterization

$$\varphi_\theta(z) = \frac{az + b}{cz + d}, \quad a, b, c, d \in \mathbb{C}, \quad ad - bc = +1, \quad (7)$$

carrying six real degrees of freedom (eight real components less the two real constraints of one complex equation; the $M \sim -M$ identification of $PSL(2, \mathbb{C})$ is discrete and removes none). In every run reported here and in the predecessor's appendices φ_θ is *frozen at the identity*; no L-BFGS-B refinement is performed. Items reach S^2 by the stereographic lift of their PCA top-2 coordinates,

$$\sigma(u, v) = \frac{1}{1 + u^2 + v^2} (2u, 2v, u^2 + v^2 - 1) \in S^2. \quad (8)$$

Fibonacci sampling and the corrected Y_3^3 constant

Sampling on S^2 uses Vogel's golden-angle lattice (Vogel 1979; Saff & Kuijlaars 1997):

$$z_i = 1 - \frac{2i+1}{N}, \quad \phi_i = 2\pi \frac{i}{\varphi_g}, \quad \theta_i = \arccos(z_i), \quad \varphi_g = \frac{1+\sqrt{5}}{2}, \quad (9)$$

so the corpus item at index i is the parameter point ($i = t$, no lookup table, $O(1)$ per point, no pole clustering). The angular probe is the three-fold azimuthal harmonic

$$Y_3^3(\theta, \phi) = K \sin^3 \theta \cos(3\phi). \quad (10)$$

The constant is corrected here. With Condon-Shortley phase (NIST DLMF),

$$K_{\mathbb{C}} = \frac{\sqrt{35}}{64\pi} = 0.417224, \quad K_{\mathbb{R}} = \frac{\sqrt{70}}{64\pi} = 0.590044, \quad (11)$$

$K_{\mathbb{R}}$ being the real-orthonormal form appropriate to a real basis. The legacy regime constant $K_{legacy} = \sqrt{245/(64\pi)} = 1.103870$ is wrong: since $245/35 = 7$ and $245/70 = 3.5$,

$$\frac{K_{legacy}}{K_{\mathbb{C}}} = \sqrt{7} = 2.645751, \quad \frac{K_{legacy}}{K_{\mathbb{R}}} = \sqrt{7/2} = 1.870829. \quad (12)$$

The error is the factor $(2\ell + 1) = 7$ counted twice — it is already inside $35/64\pi$. Every Y_3^3 magnitude published under the legacy constant is inflated by $1.870829 \times$. The correction is scale-only: because the generator of Section 5 standardizes each probe channel before scaling by the planted amplitude, switching constants leaves the emitted matrix bit-identical and V_2 identical to 10^{-12} (machine-asserted, generator self-test step 4).

Remark 1 (what φ_g and Y_3^3 cannot claim). The Fibonacci lattice is a sampling scheme and Y_3^3 a fixed basis element used as a probe; neither is a model of the corpus. Two claims are inadmissible without further work. *Azimuthal symmetry*: ϕ_i is assigned by construction from the index and the index is the PC1 rank, so power at $\cos(3\phi)$ is a property of the index map until an ordering-permutation null says otherwise. *The 384-D gate pass*: $PC1 + PC2 = 1.0000$ on the $(\ell = 384, m = 3)$ variant is a rank artifact (Proposition 2), not a fit. The golden ratio in the sampler licenses no numerology: it has no relation to any fitted κ , to pentagon geometry, or to $7/9$.

Null models and the corrected record

Throughout, $M \in \{0,1\}^{N \times D}$ is the coverage matrix and $c_i = \sum_a M_{ia}$ the row mass. The primary null is the *fixed-margin* (curveball) ensemble (Strona et al. 2014; Carstens 2015): the uniform distribution on the fibre of binary matrices with the row sums *and* column sums of M . The primary statistic is the null-standardized residual

$$\Delta V_2 z = \frac{V_2 - \mathbb{E}_0[V_2]}{SD_0[V_2]}, \quad \Delta V_2 = V_2 - \mathbb{E}_0[V_2], \quad (13)$$

with $\mathbb{E}_0, SD_0$ over the curveball ensemble at the corpus’s own N, D and marginals.

C1 — V_2 is a dimension artifact

Proposition 1 (trace-normalized share decays in D). *Let the rows be fixed and let $M^{(D)}$ append $D - D_0$ further columns to D_0 original ones. On the correlation matrix $\text{tr } \Sigma^{(D)} = D$ for every D , so $V_2^{(D)} = (\lambda_1 + \lambda_2)/D$. An appended column contributes unit trace and, unless it loads on the existing leading plane, contributes $O(1)$ to $\sum_a \lambda_a$ and $o(1)$ to $\lambda_1 + \lambda_2$. Hence $V_2^{(D)}$ is non-increasing in D up to the leading-plane loading of the new columns, with flat-spectrum floor $V_2 \rightarrow 2/D$.*

Proof sketch. z -scoring forces $\Sigma_{aa} = 1$, so $\text{tr } \Sigma = D$ and V_2 is an eigenvalue *share*, not an eigenvalue. In the limit of exact independence, adding one column adds an eigenvalue 1 to the denominator and nothing to the numerator, giving $V_2 \mapsto V_2 \cdot D_0/(D_0 + 1)$; correlated additions interpolate between that dilution and no change. For an exactly flat spectrum $\lambda_a \equiv 1$, $V_2 = 2/D$, which is therefore the iid floor. $\square$

Measured consequence, on the *identical* 2286 rows (Table 1): $V_2 = 0.7235$ at $D = 9$ and $V_2 = 0.2940$ at $D = 24$ with independent nuisance columns (0.4473 when the nuisance columns are made row-mass-correlated). The cited transition of +0.4664 from 0.2993 to 0.7657 spans the same interval, with the 0.2993 baseline the 24-D basis and the endpoint the 9-D basis. There is no transition to explain: it is Proposition 1 evaluated at two points.

basis	D	$2/D$	yubiOS	2-class	1-class (iid)	3-class	planted	GWTC-like
PCA top-2	2	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
sphere lift	3	0.6667	0.8302	0.7187	0.6843	0.6885	0.6865	0.8460
7-column subset	7	0.2857	0.7575	0.6573	0.3028	0.5923	0.3305	0.7346
9-D native	9	0.2222	0.7235	0.6454	0.2397	0.5575	0.2705	0.6624
16 real SH ($L = 3$)	16	0.1250	0.4857	0.4244	0.2202	0.3490	0.2227	0.4180
24-D indep. nuisance	24	0.0833	0.2940	0.2718	0.0984	0.2341	0.1064	0.2523
24-D mass-corr. nuis.	24	0.0833	0.4473	0.4221	0.1410	0.3620	0.1457	0.2852
384-D native (indep.)	384	0.0052	0.3828	0.2657	0.0370	0.2220	0.0358	0.0228
384-D native (replicated)	384	0.0052	0.3921	0.2567	0.0496	0.2391	0.0462	0.0451
384-D variant 1 ($\ell = 128$)	384	0.0052	0.7592	0.1569	0.0763	0.2241	0.1012	0.0594
384-D variant 2 ($\ell = 384, m = 3$)	384	0.0052	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000

Table 1. V_2 against feature-space dimension; rows held fixed within each column. Eleven bases; the 2-D and ($\ell = 384, m = 3$) bases return exactly 1.0000 on every corpus, including the single-class iid one — zero data content. The $2/D$ column is the flat-spectrum floor of Proposition 1, derived for an exactly flat spectrum only; it is not a floor for an arbitrary corpus, and several entries sit either side of it. Effective ranks quoted for this table are the proxy $\hat{r} = 2/V_2$ of Eq. (15). Source: results/ladder_VD.json.

C2 — the gate is a rank test

Proposition 2 (the gate is a rank identity under the shipped estimator). *Let*

$$PR = \frac{(\sum_a \lambda_a)^2}{\sum_a \lambda_a^2} \quad (14)$$

be the participation ratio — the quantity ordinarily meant by “effective rank”, and the one computed by the generator’s placement code (skills/curved-corpus-create/scripts/create_corpus.py). For a spectrum flat on its top- r block and negligible elsewhere, $PR = r$ and $\lambda_1 + \lambda_2 = (2/r) \sum_a \lambda_a$, so in that regime $V_2 \approx 2/PR$. The analysis code shipped with this paper (scripts/common.py) does not compute PR . It reports the two-share proxy

$$\hat{r} = \frac{2}{V_2}, \quad (15)$$

a deterministic, strictly decreasing transform of V_2 and nothing more. Under that estimator the gate equivalence

$$V_2 \geq 0.40 \Leftrightarrow \hat{r} \leq 5 \quad (16)$$

is a definitional identity: it is true by algebra, it carries no empirical content, and $\hat{r}$ adds no information to V_2 . In particular $V_2 = 1.0000 \Leftrightarrow \hat{r} = 2$, which holds exactly when the sampled design has numerical rank 2 — attainable with no data content whatsoever.

Because Eq. (16) is an identity, none of the empirical weight sits there. What is empirical is *where* the identity bites, and that is the last row of Table 1: the ($\ell = 384, m = 3, \sin^{384} \theta$) variant returns exactly 1.0000 — hence $\hat{r} = 2.00$ — on all six corpora of that table: real yubiOS, one-, two- and three-class latent mixtures, a planted-curve corpus, a GWTC-like corpus. Counting the second planted variant carried in the data file but not tabulated, results/ladder_VD.json holds 14 rows at $V_2 = 1.0000$ to machine precision, every one of them in the 2-D PCA basis or

in the $(\ell = 384, m = 3)$ variant, and none anywhere else. A single-class iid corpus — pure noise, no latent structure at all — passes the gate at exactly 1.0000 in two of the eleven bases tried. That is a fact about the bases, not about the corpora, and it is what Proposition 2 is for. Three hundred and eighty-four deterministic functions of one scalar index span a two-dimensional column space by construction; the 2-D PCA row says the same thing trivially. The predecessor’s 7-D $PC1 + PC2 = 1.0000$ on refs/ is the same artifact at the other end, its surviving coverage matrix being rank 2. Proxy values along the yubiOS ladder are $\hat{r} = 2.00$ ($D = 2$), 2.41 (3), 2.64 (7), 2.76 (9), 4.12 (16), 6.80 (24); by Eq. (15) each is $2/V_2$ and restates the yubiOS column of Table 1 rather than adding to it. A gate value of exactly 1.0000 is a red flag, not a success: the numerical rank and condition number of the design must be reported beside it, and the accept/reject decision made on $\Delta V_2 z$, which is dimension-comparable and marginal-matched.

Which effective rank is which. Two distinct quantities appear in this literature under the name “effective rank”, and every table here states which one it uses. The values in Table 1 and the 3.02 / 8.02 quoted for the GWTC-like corpus in Section 5 are the proxy $\hat{r} = 2/V_2$ of Eq. (15), computed by scripts/common.py (verified: $\hat{r} = 2/V_2$ to $\sim 10^{-15}$ on all 77 rows of ladder_VD.json). The coordinate PR in the statistic vector of Eq. (37), and the $PR = 6.837$ quoted for the worked planted corpus in Sections 5 and 6, are the participation ratio of Eq. (14), computed by create_corpus.py. The two disagree, and must: on that corpus $V_2 = 0.4477$, so $\hat{r} = 4.47$ while $PR = 6.837$. Only PR is an independent coordinate of the map; $\hat{r}$ is not, and it survives in Table 1 as a readability aid only.

C3 — the corrected null, and a retraction

quantity	value	z (real vs null)	source run
real V_2 (2286 × 9)	0.7235293730732693	—	A2-curveball-audit
published cycle-20 value	0.723529 (reproduces exactly)	—	A2-curveball-audit
curveball null , 200 × 228, 600 trades	0.709180 ± 0.001183	+12.13	A2-curveball-audit
independent swap-chain sampler	0.708593 ± 0.000507	+29.44	validation
row-marginal-only null	0.670823 ± 0.001073	+49.14	A2-curveball-audit
column-permutation null	0.239653 ± 0.002253	+214.79	A2-curveball-audit
iid Bernoulli (full entry shuffle)	0.239832 ± 0.002340	+206.70	A2-curveball-audit
row-permutation control	0.723529 ± 2.1 × 10 ⁻¹⁶ (identity check)	n/a	A-v2-nulls
Marchenko-Pastur baseline ($q = 0.003937$)	$\lambda_{\pm} = 0.878446/1.129428$; top-2 share 0.241375	—	A-v2-nulls

Table 2. Nulls for the 2286 × 9 coverage matrix. Each row is a single run, named in the last column (files under results/), and each z is recomputed from the mean and sd printed in that same row against the real $V_2 = 0.7235293730732693$; an earlier version of this table spliced mean, sd and z from different runs. The two fixed-margin samplers agree on the mean (0.709180 against 0.708593, a gap of 0.0006) and disagree on the spread: the swap chain’s sd is 2.3 × tighter, and that difference in sd is the entire difference between $z = +29.44$ and $z = +12.13$. The headline figure quoted throughout this paper is the curveball value +12.13, the more conservative of the two. The row-permutation row is an identity check — V_2 is invariant under row relabeling — not a null, and no z is defined for it.

Two independent samplers, each proved uniform on the exact row-and-column-marginal fibre, agree. Sampler A: on a 5×4 matrix whose fibre has exactly 120 elements, 300,000 draws hit all 120, hit nothing outside, and are uniform ($\chi^2 = 123.758$, $df = 119$, $p = 0.364$). Sampler B: on the exhaustively enumerated 6×4 ensemble (120 matrices, 20,000 draws), $\chi^2 = 118.096$, $df = 119$, $p = 0.506$. Marginals were asserted exactly on every one of the 2286 × 9 draws. Mixing is converged: V_2^{null} against trade count runs 0.7159258 (1N), 0.7103934 (5N), 0.7087008 (20N), 0.7092413 (100N), 0.7094042 (400N), at 12 replicates per point — flat from 20N on, within a null sd of ≈ 0.0012 . These five values are the mixing_convergence block of results/A2-curveball-audit.json verbatim; the figures printed in an earlier draft (0.715314, 0.710096, 0.708886, 0.708878, 0.709364) appear in no shipped artifact and are withdrawn — the same defect this section retracts, and it is corrected here rather than left standing. The Marchenko-Pastur prediction 0.241375 (Marchenko & Pastur 1967) lands on top of the two destroyed-dependence nulls (0.2397, 0.2398), an independent check that the pipeline is sound.

The conclusion is quantitative and modest:

$$\Delta V_2 = 0.723529 - 0.709180 = +0.014415, \quad \Delta V_2 z = +12.13, \quad \frac{\mathbb{E}_0[V_2]}{V_2} = 0.9802. \quad (17)$$

About 98% of the headline observable is fixed by row and column marginals alone; the surviving corpus-specific signal is +0.0144 of eigenvalue share. It is real ($z = +12$, not $|z| < 3$), and it is small.

Remark 2 (retraction). An intermediate internal analysis reported the same curveball null as 0.8005 ± 0.0017 , hence $z = -45.8$ — the corpus *below* its own null, “anti-structured”. That value does not reproduce and its sign is wrong. Two independent samplers, both validated as above and both converged in mixing, return 0.7089; 0.8005 is not reachable by longer mixing. Diagnostically, the same memo’s other nulls do reproduce with the present code (column-permutation claimed 0.2410, measured 0.239733 ± 0.002 ; row-mass-only claimed 0.6709, measured 0.670800 ± 0.001), which localizes the error to its curveball implementation; the memo is additionally self-inconsistent, reporting 0.7613 for the same null at the same N in a second table. Every downstream statement resting on $z = -45.8$ — in particular “the corpus is less concentrated than chance” — is withdrawn. This is the paper’s null discipline operating on its own lineage: the retraction is reported, not repaired silently.

Per-corpus residuals under the same null (50 samples each) are in Table 3. The smallest corpus, docs ($N = 126$), is indistinguishable from its own null at $z = -0.10$; the all-zero refused regime (204 identically zero rows, every column

of zero variance) has no defined V_2 and is reported as a skip, not as 0.0.

corpus	N	V_2	curveball null	z	col-perm null	MP pred.
yubiOS whole	2286	0.723529	0.7098 ± 0.0011	+12.12	0.2399	0.2414
yubiOS eligible ($c \geq 5$)	1322	0.425567	0.3824 ± 0.0051	+8.41	0.2461	0.2475
yubiOS deferred ($1 \leq c \leq 4$)	760	0.311625	0.2596 ± 0.0045	+11.65	0.2527	0.2558
yubiOS refused ($c = 0$)	204	undefined	—	—	—	—
yubiOS skills	474	0.581825	0.5559 ± 0.0122	+2.12	0.2619	0.2650
yubiOS docs	126	0.607322	0.6080 ± 0.0064	−0.10	0.2965	0.3073
yubiOS refs	791	0.648135	0.5801 ± 0.0059	+11.51	0.2532	0.2551
yubiOS self	895	0.406537	0.3863 ± 0.0035	+5.81	0.2506	0.2531
yubiOS 1391 (cycle-18)	1391	0.640256	0.5882 ± 0.0046	+11.28	0.2456	0.2469
CIS 447 (dev-sec InSpec)	447	0.331961	0.2667 ± 0.0044	+14.93	0.2635	0.2663
SCAP SSG 2107	2107	0.384629	0.3422 ± 0.0020	+21.54	0.2411	0.2422
SCAP SSG rules-dir 700	700	0.368366	0.3274 ± 0.0032	+12.92	0.2550	0.2572

Table 3. V_2 against the fixed-margin null, per corpus. Source: results/A-v2-nulls.json.

C4 — there is no critical point

The previous program located $N_c \approx 40$ as the argmax of $dV_2/d \log N$ on raw V_2 . Table 4 runs the same sweep with the null computed at each N (matched marginals, matched N , 8 subsamples per N , 120–200 null reps per point).

N	V_2 real	V_2 null (curveball)	ΔV_2	z	col-perm null
40	0.7501	0.7443	+0.0058	+0.6	0.3674
79	0.7459	0.7284	+0.0175	+2.5	0.3237
160	0.7321	0.7146	+0.0175	+3.7	0.2922
320	0.7270	0.7134	+0.0135	+4.3	0.2706
640	0.7197	0.7045	+0.0153	+6.5	0.2564
1280	0.7214	0.7069	+0.0145	+9.2	0.2462
2286	0.7235	0.7092	+0.0144	+12.3	0.2401

Table 4. Finite-size sweep with the null’s own N -dependence exhibited. ΔV_2 is flat for $N \geq 79$; the growth of z is the shrinking null SD, not a growing effect. Sources: results/finite_size.json, results/finite_size_colperm.json, results/scaling_VN.json.

Two fitted laws separate the nulls:

$$V_2^{\text{curveball}}(N) = \frac{2}{D} + 0.5252 \left(\frac{D}{N} \right)^{0.0167}, \quad V_2^{\text{colperm}}(N) = \frac{2}{D} + 0.3117 \left(\frac{D}{N} \right)^{0.5177}, \quad R_{\text{colperm}}^2 = 0.99984. \quad (18)$$

The R^2 belongs to the column-permutation fit alone: finite_size.json reports $c = 0.5252344$ and $\alpha = 0.0167256$ for the curveball law but no R^2 , and that law under-predicts the measured null by 0.008–0.010 at both ends of the sweep ($N = 40$: 0.7345 fitted against 0.7443 measured; $N = 2286$: 0.7010 against 0.7092), so it is a shape summary, not a calibrated fit. The column-permutation (iid) null relaxes toward $2/D$ like Marchenko–Pastur in the aspect ratio D/N ; the fixed-margin null relaxes toward the same asymptote only logarithmically slowly — $\alpha = 0.0167$ makes the approach invisible over the sampled range, because row masses are held — so for practical purposes it does not relax at all here. So the “ V_2 inflation at small N ” story applies to the iid ensemble only. Window slopes make the point directly: the largest $|d(\text{raw } V_2)/d \log N|$ is -0.0196 in the 79–160 window, the largest null slope is -0.0234 at 40–79, and the largest null-subtracted slope is $+0.0172$, also at 40–79 and *positive*. $\Delta V_2(N)$ has no interior maximum; the previously reported $N_c \approx 40$ was raw V_2 inheriting the null’s finite-size inflation at the smallest N in the sweep.

Generated corpora with known ground truth confirm that this flatness is the estimator working, not a blind spot. A one-class (iid) generator gives $|z| \leq 1.5$ at every N from 40 to 4096. A two-class mixture gives $\Delta V_2 = +0.0108, -0.0004, +0.0064, +0.0041, +0.0017, +0.0029, +0.0042, +0.0049$ across $N = 40, \dots, 4096$ — a constant small offset whose z climbs from $+0.9$ to $+7.7$ purely through the shrinking null SD, which is exactly the yubiOS pattern.

GL prediction (internal entry, §4)	pre-pivot status	post-null verdict	basis
P1 saturation $V_\infty \approx 0.77\text{--}0.79$	survived	reinterpreted	V_∞ is the null value, not a constant
P2 critical point at $\max dV_2/d \log N$	claimed	falsified	$\Delta V_2(N)$ flat, no interior max
P3 exponent $\beta_{emp} \approx 1/2$	not fitted	vacated	no transition to fit
P4 fluctuation peak $Var[V_2]/V_2$	not run	not tested	—
P5 susceptibility divergence $\chi(N)$	not run	not tested	—
P6 type-I/II via $\kappa = \lambda/\xi$	survived	reinterpreted	κ estimators are row-order dependent
P7 critical slowing-down	not run	not tested	—
P8 universality collapse across corpora	claimed	not supported	per-corpus z spans -0.10 to $+21.5$

Table 5. Ginzburg-Landau scoreboard. Two of eight predictions were recorded as surviving before the pivot; both are yubiOS-specific and both are reinterpreted here. The κ family is deleted by the invariance requirement of Section 6: a row-lag estimator changes under row relabeling, and the same estimator returned $\kappa \approx 1.06$ on yubiOS and $\kappa = 323$ on NIST, the latter a saturation-clamp artifact. $V_{floor} = 0.222$ is 2/9 plus finite size; $V_\infty \approx 0.769$ is an ensemble value.

The Hodge channel

The second measurement channel is combinatorial Hodge theory (Jiang et al. 2011), adopted because it supplies what an eigenvalue share cannot: a decomposition with an exact orthogonality guarantee, independent of PCA, of the stereographic lift, of λ , and of the Möbius chart.

Operators

Let $V = \{1, \dots, N\}$ be items and Λ the voters (primitives). Each voter contributes a skew-symmetric matrix and an indicator weight, aggregated as

$$Y_{ij}^\alpha = -Y_{ji}^\alpha, \quad w_{ij}^\alpha = 1 \left[\alpha \text{ compared } \{i, j\} \right], \quad w_{ij} = \sum_\alpha w_{ij}^\alpha, \quad \bar{Y}_{ij} = \frac{\sum_\alpha w_{ij}^\alpha Y_{ij}^\alpha}{\sum_\alpha w_{ij}^\alpha}. \quad (19)$$

The comparison graph is $G = (V, E)$ with $E = \{\{i, j\} : w_{ij} > 0\}$, and its 3-clique complex is $K_G^3 = (V, E, T(E))$ with

$$T(E) = \{\{i, j, k\} \in \binom{V}{3} : \{i, j\}, \{j, k\}, \{k, i\} \in E\}. \quad (20)$$

Cochain spaces: C^0 scores $s: V \rightarrow \mathbb{R}$, C^1 edge flows, C^2 triangular flows. The operators are

$$(\text{grad } s)(i, j) = (\delta_0 s)(i, j) = s_j - s_i, \quad (21)$$

$$(\text{curl } X)(i, j, k) = (\delta_1 X)(i, j, k) = X_{ij} + X_{jk} + X_{ki}, \quad (22)$$

$$(\text{div } X)(i) = -(\delta_0^* X)(i) = \sum_{j: \{i, j\} \in E} w_{ij} X_{ij}, \quad (23)$$

with the weighted inner product $\langle X, Y \rangle_w = \sum_{\{i, j\} \in E} w_{ij} X_{ij} Y_{ij}$ on C^1 and unweighted products on C^0, C^2 . Closedness $\delta_{k+1} \delta_k = 0$ gives

$$\text{curl} \circ \text{grad} = 0, \quad \text{div} \circ \text{curl}^* = 0, \quad (24)$$

and the combinatorial Laplacians are

$$\Delta_k = \delta_k^* \delta_k + \delta_{k-1} \delta_{k-1}^*, \quad \Delta_0 = \delta_0^* \delta_0 = -\text{div grad}, \quad \Delta_1 = \text{curl}^* \text{curl} - \text{grad div}, \quad (25)$$

One sign convention is used throughout, that of Jiang et al. 2011: $\Delta_0 = \delta_0^* \delta_0$ is the positive-semidefinite graph Laplacian $L = D - W$, and since Eq. (23) defines $\text{div} = -\delta_0^*$, this reads $\Delta_0 = -\text{div grad}$. Eq. (27) below ($\Delta_0 s = -\text{div } \bar{Y}$, $s^* = -\Delta_0^\dagger \text{div } \bar{Y}$) and the expression for Δ_1 are stated in that same convention. Δ_1 is the graph Helmholtzian, with $\dim \ker \Delta_k = \beta_k(K)$. The decomposition is

$$C^1 = \underbrace{\text{im}(\text{grad})}_{\text{globally consistent}} \oplus \underbrace{\ker(\Delta_1)}_{\text{harmonic: globally cyclic}} \oplus \underbrace{\text{im}(\text{curl}^*)}_{\text{locally cyclic}}, \quad \ker \Delta_1 = \ker(\text{curl}) \cap \ker(\text{div}), \quad (26)$$

the three summands mutually orthogonal in $(\cdot, \cdot)_w$. The least-squares potential and the residual split are

$$\min_{s \in C^0} \|\text{grad } s - \bar{Y}\|_{2,w}^2 \Leftrightarrow \Delta_0 s = -\text{div } \bar{Y}, \quad s^* = -\Delta_0^\dagger \text{div } \bar{Y}, \quad (27)$$

$$R^* = \bar{Y} - \text{grad } s^*, \quad \Phi^* = \left(\text{curl} \text{curl}^* \right)^\dagger \text{curl } \bar{Y}, \quad \text{proj}_{\ker \Delta_1} = I - \Delta_1^\dagger \Delta_1. \quad (28)$$

Orthogonality turns the split into a genuine energy budget:

$$\|\bar{Y}\|^2 = \|\bar{Y}_{grad}\|^2 + \|\bar{Y}_{curl}\|^2 + \|\bar{Y}_{harm}\|^2, \quad \rho_g = \frac{\|\bar{Y}_{grad}\|^2}{\|\bar{Y}\|^2}, \quad \rho_c = \frac{\|\bar{Y}_{curl}\|^2}{\|\bar{Y}\|^2}, \quad \rho_h = \frac{\|\bar{Y}_{harm}\|^2}{\|\bar{Y}\|^2}, \quad (29)$$

with $\rho_g + \rho_c + \rho_h = 1$. Betti bookkeeping, with $B_1 \in \mathbb{R}^{|E| \times |V|}$ the matrix of δ_0 and $B_2 \in \mathbb{R}^{|T| \times |E|}$ that of δ_1 :

$$\beta_0 = |V| - \text{rank } B_1, \quad \beta_1 = |E| - \text{rank } B_1 - \text{rank } B_2, \quad \beta_2 = |T| - \text{rank } B_2, \quad (30)$$

checked against $\chi = |V| - |E| + |T| = \beta_0 - \beta_1 + \beta_2$. The flow is Construction A1, the Jaccard-normalized dominance

$$\bar{Y}_{ij}^{A1} = \frac{1}{|S_{ij}|} \sum_{a \in S_{ij}} (M_{ja} - M_{ia}) = \frac{c_j - c_i}{|\mathbf{m}_i \vee \mathbf{m}_j|}, \quad S_{ij} = \{a: M_{ia} + M_{ja} \geq 1\}, \quad (31)$$

whose pair-dependent denominator is what stops $\bar{Y}$ from being a pure gradient: with all primitives voting on every pair, $\bar{Y}_{ij} = (c_j - c_i)/d$ is exactly a gradient and $\rho_c = \rho_h = 0$ identically. Finally, the discrete circulation of a cycle $\sigma = (i_1 \rightarrow \dots \rightarrow i_L \rightarrow i_1)$,

$$W(\sigma) = \sum_{l=1}^L X_{i_l i_{l+1}}, \quad (32)$$

vanishes on gradients, is contractible on curls, and on $\ker \Delta_1$ depends only on the homology class of σ — a real-valued circulation class, not a quantized topological charge.

Measurements

On yubiOS ($N = 2286$; symmetrized k -NN with $k = 8$ under Jaccard similarity; seed 20260812): $|E| = 16,438$, $|T| = 13,799$, and only 176 distinct rows of 512 possible. Measured,

$$\rho_g = 0.99961366, \quad \rho_c = 0.00013059, \quad \rho_h = 0.00025575, \quad (33)$$

with $\beta_0 = 6$, $\beta_1 = 6756$, $\beta_2 = 6397$, rank $B_1 = 2280$, rank $B_2 = 7402$, and the Euler identity exact ($\chi = -353 = -353$).

Numerical hygiene: $(\text{grad}, \text{curl})/\|Y\|^2 = -5.4 \times 10^{-20}$, $(\text{grad}, \text{harm})/\|Y\|^2 = -6.9 \times 10^{-16}$, $(\text{curl}, \text{harm})/\|Y\|^2 = -1.1 \times 10^{-20}$, $\rho_g + \rho_c + \rho_h - 1 = 1.3 \times 10^{-15}$, $B_2 B_1$ exactly zero, and a hexagon control returning $\beta_1 = 1$ — the last of these run out-of-band, since `hodge_decompose.py` exposes no callable hexagon entry point and `proofs/VERIFY.sh` substitutes a 20-node smoke decomposition in its place (`proofs/PROOFS.md`). On the cycle-18 corpus ($N = 1391$, same construction): $|E| = 10,236$, $|T| = 7149$, $\rho_g = 0.997422$, $\rho_c = 0.000841$, $\rho_h = 0.001737$, $\beta_1 = 5293$.

statistic	real	curveball- M null ($n = 50$)	z	degree-rewired G ($n = 50$)	z
ρ_{grad}	0.999614	0.999636 ± 0.000077	-0.29	0.974942 ± 0.000418	+59.04
ρ_{curl}	0.000131	0.000181 ± 0.000044	-1.12	0.001038 ± 0.000114	-7.98
ρ_{harm}	0.000256	0.000183 ± 0.000051	+1.39	0.024020 ± 0.000447	-53.14
β_1	6756	6561.7 ± 56.9	+3.41	$13,677.9 \pm 22.1$	-312.54

Table 6. Hodge quantities against two nulls: the matrix null (curveball on M , k -NN graph rebuilt from each drawn matrix) and the graph null (same M , same degrees, edges rewired). Every z is recomputed as $(\text{real} - \text{null mean})/\text{null sd}$ from the values printed in its own row, and each is attributed to its own ensemble: the +3.41 on β_1 is against the curveball- M null, the -312.54 against the degree-rewired null. One caution on the matrix-null column: because it rebuilds the graph per draw, its own real baseline is the rebuilt graph ($|E| = 16,465$, $\beta_1 = 6702$, $\rho_h = 0.00025453$) rather than the primary run's ($|E| = 16,438$, $\beta_1 = 6756$, $\rho_h = 0.00025575$) — a shift of about 0.2%. Against that rebuilt baseline the stored z for β_1 is +2.47; against the primary $\beta_1 = 6756$ printed in the “real” column here it is +3.41, and +3.41 is the figure used in the text. Source: `results/B-hodge-yubios.json`.

The reading is sharp. Against the *matrix* null every Hodge quantity sits within $\sim 2.5\sigma$: the coverage matrix's own marginals explain its entire Hodge profile. Against the *graph* null the corpus is dramatically less cyclic and less holey than chance. **The corpus is strongly anti-cyclic — it has far fewer independent holes than a random graph of the same shape.** This inverts rather than supports the Ginzburg-Landau vortex reading: harmonic mass is not concentrating in defect cores; there is less of it than chance, and it avoids the defect rows entirely.

Three pre-registered predictions, all falsified

P-H1 (“ ρ_h collapses across $N_c \approx 40$ -200, beyond the graph null”) fails on its own stated criterion. With $k = 8$ fixed and 20 independent subsamples per N , ρ_h rises from 0.001405 ± 0.000270 at $N = 40$ to 0.003567 ± 0.000805 at $N = 80$ (a $+2.5\sigma$ increase), and at $N = 160$ (0.002412) is still above the $N = 40$ value. The monotone decline begins only after $N \approx 320$ and tracks edge density ($0.042 \rightarrow 0.0063$) rather than any critical band. What survives is the mechanism check, not the phenomenon: the rewired null rises monotonically ($0.0014 \rightarrow 0.0240$) where the real corpus falls, so the separation is real and enormous (z from $+0.04$ at $N = 40$ to -353.3 at $N = 2286$) — but it is smooth density-tracking, not a transition at N_c . This also removes the load-bearing status the pivot memo had assigned to P-H1's outcome.

P-H2 (“ ρ_c tracks $1 - V_2$ with Spearman $\rho_s \geq 0.6$ ”) fails with the wrong sign: across 11 corpora and subcorpora, $\text{Spearman}(1 - V_2, \rho_c) = -0.4545$, $p = 0.160$, and $\text{Spearman}(1 - V_2, \rho_h) = -0.4727$, $p = 0.142$. The curl maximum is docs ($N = 126$, $\rho_c = 0.011903$), the *smallest* corpus: ρ_c tracks graph density far more visibly than it tracks $1 - V_2$. The prediction's second clause (the 24-D variant should be the curl maximum) is untestable on the shipped data, which contains 9-D matrices only, and is reported as not-tested.

P-H3 (“defect rows carry harmonic support, $\text{lift} \geq 2$ ”) fails by more than a factor of ten. With E_D the defect-incident edges, $\text{lift} = \left(\|h\|_{E_D}^2 / \|h\|^2 \right) / \left(|E_D|/|E| \right)$ is exactly 0.0 on the $k = 8$, $k = 16$ and $\tau = 0.35$ graphs (and 0.0 on the 1391-row corpus with its 14 strict defects), and 0.2054 on the coverage-band graph, against a calibration value of 0.17 for *randomly placed* zero rows. The mechanism is plain: the 204 zero-coverage rows are mutually identical, so k -NN links them only to each other, where $|\mathbf{m}_i \mathbf{v} \mathbf{m}_j| = 0$ and the flow is identically zero — they form isolated

components ($\beta_0 = 6$ at $k = 8$, 221 on the threshold graph). Clause (ii), quantized azimuthal S^2 winding, was *not run* — the shipped inputs carry no S^2 embedding coordinates — and is reported as untested, not as a pass.

Reporting three falsifications from three pre-registered predictions is the methodology working, not the channel failing. The falsifications are what license the replacement claim (anti-cyclicity), which was not predicted in advance and would have been unavailable without the graph null.

Construction invariance

construction	$ E $	β_0	β_1	ρ_g	ρ_c	ρ_h	$z(\rho_c)$	$z(\rho_h)$	$z(\beta_1)$
k -NN $k = 8$	16,438	6	6756	0.999614	0.000131	0.000256	-7.98	-53.14	-312.5
k -NN $k = 16$	32,068	2	7222	0.998833	0.000542	0.000624	-22.09	-62.68	-298.8
threshold $\tau = 0.35$	20,000	221	13,286	0.999441	0.000141	0.000418	-1.91	-19.12	-16.1
coverage-band $b = 1$	20,000	1	14,886	0.990753	0.001099	0.008148	+1.29	-26.34	-37.3

Table 7. Four graph constructions on the same M , each against its own degree-rewired null. The $\tau = 0.35$ and coverage-band rows both sit exactly at the builder’s $|E| = 20,000$ edge cap (field edge_cap in the source JSON; the two k -NN rows are uncapped at 60,000), so their β_0 and β_1 are in part cap artifacts — which is worth stating, since these two rows carry the $32 \times$ spread in ρ_h and the sign flip of the ρ_c verdict. Source: results/D-construction-sensitivity.json.

Construction-invariant, and therefore reportable as corpus properties: (i) $\rho_g > 0.99$ everywhere — the flow is dominated by a potential and the corpus is near-totally ordered by coverage; (ii) ρ_h far below the degree-matched null ($z \in [-63, -19]$); (iii) β_1 far below it ($z \in [-313, -16]$); (iv) defect lift $\ll 2$ (max 0.205); (v) orthogonality residuals $< 10^{-15}$ and the Euler identity satisfied in every run. *Not invariant*, and therefore not reportable as corpus scalars: the magnitude of ρ_h (spans $32 \times$, $0.000256 \rightarrow 0.008148$), of β_1 ($2.2 \times$) and β_0 ($1 \rightarrow 221$); the sign of the ρ_c verdict (below null at k -NN, indistinguishable at threshold, above null at coverage-band); and harmonic-support concentration (26 edges carry 50% and 369 carry 90% of harmonic energy at k -NN, of 16,438, with only 712 of 14,158 fundamental cycles carrying non-zero circulation, whereas on the coverage-band graph the support is diffuse — 672 edges for 50% and 12,623 of 17,715 cycles non-zero).

The standard-candle generator and calibrated detection

Everything above is negative or corrective. The positive result is an instrument: a generator of corpora whose curvature is prescribed by construction, so that a detection statistic can be calibrated rather than asserted.

Generator

Fix N , d , a base rate p_0 , a mode count m , and an amplitude $s \geq 0$. Lay down the Fibonacci lattice of Eq. (9), evaluate the 16 real spherical harmonics of $L = 3$ at each point x_i , and take the first m probe channels $g_1, \dots, g_m$ in the fixed order $Y_3^3, Y_3^{-3}, Y_2^2, Y_2^{-2}, Y_3^0$ (channel 0 is always Y_3^3). Standardize each channel,

$$\tilde{g}_k(x_i) = \frac{g_k(x_i) - \bar{g}_k}{sd(g_k)}, \quad (34)$$

which is what makes the construction invariant to the normalization constant K of Eq. (11). Assign deterministic column loadings w_{kj} , column j carrying channel $j \bmod m$ with sign alternating by block, so that distinct columns co-vary *through the planted curve* rather than merely through row mass. Then

$$\eta_{ij} = \mu + s \sum_{k=1}^m \tilde{g}_k(x_i) w_{kj}, \quad \mu = \text{logit}(p_0), \quad (35)$$

$$X_{ij} \sim \text{Bernoulli}(\sigma(\eta_{ij})), \quad \sigma(u) = \frac{1}{1 + e^{-u}}. \quad (36)$$

The amplitude s , in log-odds units, is the ground truth. Two further kinds are generated for reference: mixture, a k -class latent mixture with random column-probability centres, and null, iid Bernoulli at the column marginals of the matched $s = 0$ corpus.

Detection

Procedure 1 (calibrated detection).

Given a binary corpus X of shape $N \times d$: (1) compute V_2 by Eq. (5); (2) draw R matrices from the curveball ensemble at X ’s own N , d and marginals; (3) report $\Delta V_2 z$ by Eq. (13) *together with* R , since z is rep-count sensitive at the 15–20% level; (4) reject the matched-marginal null M_0 when $\Delta V_2 z > 1.645$. Procedure 1 is a definition, not a result. What follows is its measured behaviour on corpora whose amplitude s is known by construction — an empirical characterization on one generator family and one grid, with no claim of generality beyond it.

Empirical Result 1 (measured detection behaviour). Let X be generated by Eqs. (34)–(36) at amplitude s and passed through Procedure 1. Over the grid of Table 8 — one generator family, one base rate, $m = 2$ modes, 8

independent corpora per cell at $N \in \{128, 512\}$ and 4 at $N = 2048$, each against its own curveball ensemble — the following is measured, not proved:

- *Detection power is monotone non-decreasing in s over $s \geq 0.5$ at every (N, d) in the grid, and strictly increasing for $N \geq 512$.* At $N = 512$, $d = 9$ the power at $z > 1.645$ runs $0.00 \rightarrow 1.00 \rightarrow 1.00$ across $s = 0.5, 1, 2$; at $N = 512$, $d = 16$, $0.25 \rightarrow 1.00 \rightarrow 1.00$; at $N = 2048$, $d = 9$, 1.00 throughout; at $N = 2048$, $d = 16$, $0.50 \rightarrow 1.00 \rightarrow 1.00$. Cell-mean ΔV_{2z} is likewise increasing in s over the same range at $N \geq 512$ ($+0.04 \rightarrow +3.43 \rightarrow +12.86$ at $N = 512$, $d = 9$; $+2.46 \rightarrow +10.82 \rightarrow +28.07$ at $N = 2048$, $d = 9$), and increasing in N at fixed $s \geq 1$ ($+0.95, +3.43, +10.82$ at $s = 1$, $d = 9$, $N = 128, 512, 2048$).
- *At $s = 0$ the statistic is centred at zero and over-dispersed, and the false-positive rate is bounded but above nominal.* On 48 fresh $s = 0$ corpora at $N = 512$, $d = 9$: mean $z = -0.014$, sd 1.320 where a calibrated z would give 1.000 ; one-sided $FPR = 0.083$ at the nominal-5% threshold $z > 1.645$, two-sided 0.104 at $|z| > 1.96$. The over-dispersion is disclosed here and carried through Section 7: every z in this paper should be read as inflated by roughly a third in units of sd.
- *Monotonicity fails at low power, and no ordering is claimed there.* At $N = 128$, $d = 9$ the cell means are $+0.59 \pm 0.36$ ($s = 0$), $+0.63 \pm 0.33$ (0.25), $+0.09 \pm 0.65$ (0.5), $+0.95 \pm 0.66$ (1.0) — unordered, and every one within Monte-Carlo error of zero at 8 reps per cell ($\pm$ is the standard error of the cell mean). The same happens at $s = 0.25$ wherever power sits on its floor: -1.04 ± 0.42 at $N = 2048$, $d = 9$, below the $+0.32 \pm 0.53$ of $s = 0$ in the same column, and at $N = 512$, $d = 9$ the $s = 0.25$ cell records power 0.12 against 0.00 at $s = 0.5$ — one of eight corpora crossing threshold by chance, which is what an 8.3% false-positive rate looks like at eight replicates. Cell means at $s \leq 0.25$, and all cell means at $N = 128$, must not be read as ordered.
- *The sensitivity limit is $s \approx 1$ log-odds at $N \leq 512$.* At $N = 512$, $d = 9$: $s = 0.5$ gives $+0.04$ and power 0.00 ; $s = 1.0$ gives $+3.43$ and power 1.00 . A non-detection below that amplitude is a statement about power, not about the world.
- *Raw V_2 has none of these properties:* it orders corpora by (N, d) rather than by s . $V_2 = 0.3083$ at $(N = 128, d = 9, s = 0)$ exceeds $V_2 = 0.3077$ at $(N = 2048, d = 9, s = 2)$, while ΔV_{2z} separates the same pair as $+0.59$ against $+28.07$.

None of the above is a theorem, and it is deliberately not presented as one: there is no hypothesis set that survives outside this grid, every clause is design-dependent, and the thresholds do not transfer across N , d , base rate, or mode count (Section 7). Presenting a Monte-Carlo sweep in a theorem environment is precisely the failure this paper indicts elsewhere in its own lineage. All numbers in (i)–(v) are from results/signal_recovery.json (cell means and standard errors recomputed from its 248 per-corpus rows) and results/SUMMARY.md §2.

N	d	s	reps	V_2	V_2 null	ΔV_2	ΔV_{2z}	power	R_{paper}^2	R_{cv}^2
128	9	0.0	8	0.3083	0.3019	+0.0064	+0.59	0.12	1.0000	0.0568
128	9	0.25	8	0.3079	0.3004	+0.0076	+0.63	0.25	1.0000	0.0487
128	9	0.5	8	0.3010	0.3003	+0.0007	+0.09	0.25	1.0000	0.0554
128	9	1.0	8	0.3098	0.3002	+0.0096	+0.95	0.25	1.0000	0.1354
128	9	2.0	8	0.3428	0.3013	+0.0415	+3.69	0.88	1.0000	0.0944
128	16	2.0	8	0.2361	0.1984	+0.0377	+5.34	1.00	1.0000	0.0612
512	9	0.0	8	0.2587	0.2603	−0.0016	−0.27	0.00	1.0000	0.0928
512	9	0.5	8	0.2605	0.2604	+0.0001	+0.04	0.00	1.0000	0.1063
512	9	1.0	8	0.2783	0.2615	+0.0168	+3.43	1.00	1.0000	0.0973
512	9	2.0	8	0.3250	0.2613	+0.0637	+12.86	1.00	1.0000	0.2033
512	16	1.0	8	0.1811	0.1602	+0.0209	+6.61	1.00	1.0000	0.0940
512	16	2.0	8	0.2236	0.1612	+0.0624	+20.32	1.00	1.0000	0.1432
2048	9	0.0	4	0.2415	0.2407	+0.0007	+0.32	0.00	1.0000	0.1152
2048	9	0.5	4	0.2469	0.2412	+0.0057	+2.46	1.00	1.0000	0.1067
2048	9	1.0	4	0.2712	0.2411	+0.0301	+10.82	1.00	1.0000	0.1010
2048	9	2.0	4	0.3077	0.2423	+0.0654	+28.07	1.00	1.0000	0.1702
2048	16	1.0	4	0.1689	0.1422	+0.0267	+18.68	1.00	1.0000	0.1133
2048	16	2.0	4	0.2266	0.1428	+0.0838	+57.67	1.00	1.0000	0.1942

Table 8. Signal recovery and power (selected rows of the 30-cell sweep; “power” is the detection rate at $z > 1.645$). R_{paper}^2 is the predecessor’s sphere-fit statistic: 1.0000 in every cell, at every amplitude, including $s = 0$. Sources: results/signal_recovery.json, results/SUMMARY.md §2.

The sphere-fit R^2 is degenerate; its cross-validated form is not

The last two columns of Table 8 carry a second correction. The predecessor’s sphere-fit R^2 returns 1.0000 on everything — planted corpora at every amplitude, and independent-column noise — because the $L = 1$ spherical-

harmonic block is linear in (x, y, z) and the fit is evaluated in-sample on points constructed from the very embedding it is fitting. It is not a measure of curvature; it is an identity. The cross-validated $L \geq 2$ variant is informative: on a GWTC-like corpus at $N = 1000$ it returns $R_{cv}^2 = 0.7678$ against 0.1391 on that corpus's column-permuted null (at $N = 138$: 0.6148 against 0.0175). The same pair in V_2 : 0.6624 with $z = +111.3$ and proxy $\hat{r} = 3.02$ for the real generator, against 0.2495 with $z = -0.13$ and $\hat{r} = 8.02$ for its null (both $\hat{r} = 2/V_2$, Eq. (15)). The false-positive block confirms that $z(\Delta R_{cv}^2)$ is conservative rather than anti-conservative: mean -0.265 , sd 0.884 , one-sided $FPR = 0.021$.

The generator's own calibration pack ($N = 200$, $d = 9$, $m = 2$, 8 trials per amplitude, 60 null reps) separates statistic from diagnostic in one table: $\Delta V_2 z$ sweeps $+0.18, +0.10, +1.50, +5.49, +10.28, +19.34, +28.01$ across amplitudes $0, 0.25, 0.5, 0.75, 1.0, 1.5, 2.0$, reaching power 1.00 at $s = 0.75$, while the sphere R^2 stays flat at ≈ 0.48 across the entire sweep and the gate $PC1 + PC2$ passes throughout. Detection threshold: $s = 0.75$ at this (N, d, p_0, m) ; FPR at $|z| > 3$ is 0.00, at $|z| > 2$ is 0.125. A worked single corpus at $s = 1.5$: $V_2 = 0.4477$, participation ratio $PR = 6.837$ (Eq. (14); the two-share proxy on the same corpus is $\hat{r} = 2/0.4477 = 4.47$), sphere $R^2 = 0.4330$, $PC1 + PC2 = 0.4474$ (gate passes), design numerical rank 16, condition number 1.006, curveball null 0.2864 ± 0.0091 , $\Delta V_2 = +0.1613$, $\Delta V_2 z = +17.67$ at 200 null reps. The gate passing at 0.4474 says little; the load-bearing line is $+17.67$. The same corpus is measured again in Section 6 at 60 null reps and scores $+20.62$: z is rep-count sensitive at the 15–20% level, which is why Procedure 1 requires the rep count to be reported beside every z .

The IS-THIS-X question space

Definition

Definition 1 (question space). The question space is the triple $(\Phi, \mathcal{M}, \mathcal{V})$, where Φ maps a corpus to a null-standardized statistic vector, $\mathcal{M}$ is an extensible set of reference families generated at the observed $(N, d, \text{density})$, and $\mathcal{V}$ is an exclusion-only verdict rule.

The statistic vector is

$$s(M) = [V_2, \Delta V_2 z, PR, Otsu(\text{row-mass histogram}), \bar{\pi}, sd(\pi)], \quad \pi a = \frac{1}{N} \sum_i M_{ia}, \quad (37)$$

with PR the participation ratio of Eq. (14) — not the two-share proxy $\hat{r}$, which would be a deterministic function of the first coordinate and would add a spurious dimension to the question space — optionally extended by the Hodge block $[\rho_g, \rho_c, \rho_h, \beta_1]$ under a fixed, declared graph construction. Every coordinate is invariant under row relabeling; the marginal coordinates are equivariant under column permutation and are summarized by permutation-invariant moments. The invariance requirement is not decoration: it deletes the entire row-lag κ family by construction, since a statistic computed along file-listing order cannot be a property of a corpus.

Each coordinate is standardized against each family's own mean and sd, with a 5% relative SD floor so a noisily narrow family cannot exclude its own members:

$$z_{f,c}(M) = \frac{s_c(M) - \mathbb{E}_f[s_c]}{\max(SD_f[s_c], 0.05|\mathbb{E}_f[s_c]|)}. \quad (38)$$

The verdict rule is exclusion-only:

$$\mathcal{V}(M, f) = \begin{cases} \text{excluded} & \text{if } \max_c |z_{f,c}(M)| > 3, \\ \text{not-excluded} & \text{if } \max_c |z_{f,c}(M)| \leq 3, \\ \text{not-tested} & \text{if } f \text{ is degenerate on every coordinate.} \end{cases} \quad (39)$$

PARTIAL and “compatible with” are not permitted outputs. The default families are M_0 (iid Bernoulli at matched column marginals — the origin), M_1^{weak} and M_1^{strong} (planted curve at $s = 0.5$ and $s = 1.5$), and M_4 (two-class latent mixture at separation 1.5). The set is extensible, and the correct response to an all-excluded result is to widen the grid, not to relabel the nearest family.

Round-trip validation

The map is trustworthy only if a corpus of known provenance recovers its own family and excludes the origin. On the planted corpus at $N = 200$, $d = 9$, $s = 1.5$ — the same corpus worked through in Section 5, here measured at 60 null reps rather than 200, hence $\Delta V_2 z = 20.62$ against the $+17.67$ quoted there — the measured vector is

$$s = [V_2 0.4477, \Delta V_2 z 20.62, PR 6.837, Otsu 0.6976, \bar{\pi} 0.5139, sd(\pi) 0.0343], \quad (40)$$

with verdicts: M_0 excluded at $\max |z| = 27.91$ (driving coordinate $\Delta V_2 z$); M_1^{weak} excluded at 15.15 ($\Delta V_2 z$); M_1^{strong} not-excluded at $\max |z| = 0.99$ (driving coordinate PR); M_4 excluded at 3.89, driven by column-marginal

spread rather than by V_2 (where $z = 2.07$) — which is precisely the coordinate on which curve and mixture genuinely overlap. A matched null corpus returns $\Delta V_2 z = -0.116$ and does not exclude M_0 . Own family recovered at $\max |z| < 1$, origin excluded at $z \approx 28$: that round trip is what licenses reading the map on real corpora.

Placement of yubiOS and of the GWTC-like corpus

family	driving coordinate	evidence	verdict for yubiOS (2286 × 9)
M_0 matched-marginal null	$\Delta V_2 z$	+12.13	excluded
M_2 mean-field Ginzburg-Landau	shape of $\Delta V_2(N)$	flat, no interior max	excluded
M_4 two-population mixture	V_2, PR , Otsu	nearest family	not-excluded
M_6 Hodge / topological (cyclicity)	ρ_h, β_1 vs matrix null	+1.39, +3.41	not-excluded (no excess cyclicity)
M_1 planted curve	$\Delta V_2 z$ magnitude	+12 vs +20 to +28 at $s \geq 1.5$	not-tested at matched (N, d)

Table 9. Placement of yubiOS. The corpus sits off the origin, but only just: the nearest family is a two-population mixture, with a small positive residual $\Delta V_2 = +0.0144$.

yubiOS is M_0 -excluded at $z = +12.3$, nearest a two-population latent-class mixture with a small positive residual. That one sentence subsumes the cross-corpus work of cycles 27–34: corpora with unimodal row mass sit near the origin (docs, $z = -0.10$), and corpora that pool a saturated population with a sparse one sit far from it (whole corpus $z = +12.1$; SCAP SSG $z = +21.5$). Mean-field Ginzburg-Landau is excluded on the shape of $\Delta V_2(N)$ (Table 4), not on the sign of a single z — which matters, because the sign of that single z is exactly what the retraction of Remark 2 reversed. The generated latent-class series reproduces yubiOS’s signature: a two-class mixture shows a small constant ΔV_2 whose z grows only with N (+0.9 at $N = 40$ to +7.7 at $N = 4096$), while a one-class corpus stays at $|z| \leq 1.5$ throughout.

The GWTC-like corpus places elsewhere, and is the useful contrast: $V_2 = 0.6624$, $\Delta V_2 z = +111.3$, proxy $\hat{r} = 3.02$, $R_{cv}^2 = 0.7678$ at $N = 1000$, against its own column-permuted null at $V_2 = 0.2495$, $z = -0.13$, $R_{cv}^2 = 0.1391$; at $N = 138$, $V_2 = 0.6722$, $z = +34.3$. The two corpora are not near neighbours in s -space — which is the honest form of the cross-domain claim earlier cycles tried to make by narrative. It must also be said that a continuous physical-parameter matrix and a binary coverage matrix are not the same object: the GWTC-like corpus enters this map only through a stated binarization, and any claim about real gravitational-wave catalogues would need real posterior samples under that same stated rule.

Scope and honest caveats

The retraction is part of the record. Section 3.3 withdraws 0.8005 / $z = -45.8$ after two independent replications. The correct posture toward the numbers replacing it is identical: 0.709180 ± 0.001183 is reproducible from the shipped scripts and seed, and should be re-run rather than cited.

z is over-dispersed. The calibration block measures $sd[z(\Delta V_2)] = 1.320$ at $s = 0$ where a perfectly calibrated z would give 1.000, with a one-sided FPR of 8.3% at the nominal 5% threshold. Every z here should be read as inflated by roughly a third in units of sd . This does not touch conclusions at $|z| > 12$; it does mean that z between 2 and 4 — the +2.12 for skills, the +3.41 for β_1 against the matrix null, the +2.5 at $N = 79$ — must not be treated as decisive.

Hodge magnitudes are construction-dependent. Only the directional statements of Table 7 are corpus properties. ρ_h spans $32 \times$ and β_1 spans $2.2 \times$ across four defensible constructions of the same matrix, and the sign of the ρ_c verdict flips. Reporting $\beta_1 = 6756$ as a corpus scalar would be an artifact; reporting “ β_1 sits 16–313 σ below its degree-matched null in every construction tried” is not. Separately, rank B_2 is computed by GF(2) elimination on the large graphs; it agreed with exact real rank wherever a dense SVD was feasible (synthetic $N = 300$: $991 = 991$; coverage-band graph: $2829 = 2829$), but 2-torsion in the clique complex would break that agreement and could not be excluded on the k -NN and threshold graphs. Graph-construction artifacts that mimic transitions — the Kahle density window — remain the first-class risk of this channel, which is why every Hodge number here carries a degree-matched null.

The sensitivity floor is $s \approx 1$ log-odds. At $N = 512$, $d = 9$, amplitude $s = 0.5$ gives $\Delta V_2 z = +0.04$ and power 0.00; $s = 1.0$ gives +3.43 and power 1.00. A non-detection at these N is a statement about power, not about the world, and must be reported as one. The generator’s own threshold at $N = 200$ is $s = 0.75$, and thresholds do not transfer across N , d , base rate, or mode count.

What the Hodge nulls do not settle. Against the matrix null every Hodge quantity is within $\sim 2.5\sigma$, so the honest summary is that the coverage matrix’s marginals plus the chosen graph jointly determine the entire Hodge profile. Anti-cyclicity is a statement about which items are near which items, not about latent capability organization beyond marginals. The Ginzburg-Landau bridge should be retired on this evidence: there is no free energy, no dynamics, no ξ , no λ ; the Hodge channel does not repair κ , and ρ_c is a disorder measure, not a temperature.

Untested items, named. P-H3 clause (ii), quantized azimuthal S^2 winding — no S^2 coordinates in the shipped inputs. P-H2’s 24-D clause — no 24-D matrix in the shipped data. GL predictions P4, P5, P7 (fluctuation peak, susceptibility, critical slowing-down) — not run. The ordering-permutation null required by Remark 1 for any azimuthal claim — specified, not yet executed on the real corpus. Dynamics: the edit log exists (1391 atom dispatches with per-

cycle coverage vectors) and a detailed-balance test on it remains the one clean way to separate potential from non-potential dynamics; until it is run, “emergence over cycles” should be restated as an append-only policy saturating a bounded coverage space.

What is inherited and not re-verified. The predecessor’s atom-level results (the empirical non-negativity of Δ over 1391 + 124 dispatches with zero violations; the k -quantized Δ ladder peaking at $k = 5$) are not re-examined here, nor is the matched-parameter ablation, which remains two single-seed point estimates without error bars. Nothing in this paper contradicts them; nothing strengthens them either. Finally, the corpus’s own row-mass bimodality is partly manufactured by the improvement loop that produced it — an append-only policy saturating skills, docs and refs while self stayed sparse — which is a plausible mechanism for the mixture placement of Section 6 but has not been traced row by row and is flagged as unconfirmed.

Conclusion

The program’s headline observable is, to 98%, a function of its own marginals. Its celebrated transition is a change of basis on identical rows (Proposition 1). Its gate is a rank test that a corpus of pure noise passes at exactly 1.0000 in two of the eleven bases of Table 1 (Proposition 2), and the “effective rank” reported beside it is $2/V_2$, a restatement of the gate rather than a check on it. Its critical point disappears when the null’s own finite-size behaviour is subtracted. Its Hodge reading is inverted: the corpus is anti-cyclic, with β_1 some 313σ below a degree-matched null, and all three pre-registered predictions of that channel are falsified. And its most-cited intermediate correction — the $z = -45.8$ reversal — was itself wrong and is retracted here on the strength of two independently validated uniform samplers.

What survives, measured rather than asserted, is a residual of $\Delta V_2 = +0.0144$ at $z = +12.3$: small, real, and placed. And what replaces the sequence of candidate X ’s is the map of Eq. (1) together with the instrument that calibrates it — a generator whose planted amplitude is known, on which $\Delta V_2 z$ has power monotone in signal above $s = 0.5$, is quiet (though over-dispersed) at zero signal, and powered to 1.00 at one log-odds by $N = 512$, while the statistics the program previously reported (V_2 , the gate, the sphere-fit R^2) are flat, degenerate, or anti-ordered over the same sweep. yubiOS is then not “partially compatible with Ginzburg-Landau”. It is: M_0 excluded at $z = +12.3$, mean-field GL excluded on the shape of $\Delta V_2(N)$, nearest family a two-population mixture, no excess cyclicity, off-map families named as not-tested rather than as partial. The next corpus can be placed on the same map by the same script, and the map’s own round trip — planted excludes the null at $z = 27.9$, recovers its own family at $\max |z| = 0.99$ — is what makes that placement mean anything.

Appendix A: Reproduction manifest

All paths are relative to the evidence bundle root. Master seed 20260812; Python 3.12.12, numpy 2.5.1, scipy 1.18.0. Null convention: fixed-margin curveball (10 N trades unless stated; the headline V_2 null uses 200 samples $\times$ 228,600 trades) for binary matrices, independent column permutation for continuous ones.

Scripts (scripts/).

common.py (shared loaders, V_2 , the two-share proxy $\hat{r} = 2/V_2$ of Eq. (15), spectra); null_models.py (curveball, swap-chain, column permutation, iid, row-mass-only; sampler uniformity harness); hodge_decompose.py (Eqs. (21)–(30): boundary matrices, LSMR solves, ρ_* , GF(2) Betti numbers, orthogonality assertions, hexagon control); run_real_corpora.py (per-corpus V_2 /Hodge against both nulls); audit_curveball_discrepancy.py (the retraction audit of Remark 2: fibre enumeration, χ^2 uniformity, mixing sweep, cross-checks against the other three nulls); generate_samples.py (planted / mixture / null corpora, Eqs. (9), (34)–(36)); ladder_correlations.py (Table 1); timeseries_build.py (consolidated series).

Results (results/).

A-v2-nulls.json (Table 3); A2-curveball-audit.json (Table 2, mixing sweep, retraction evidence); validation.json and validation-curveball-sampler.json (sampler uniformity: $\chi^2 = 118.096$, $p = 0.506$; $\chi^2 = 123.758$, $p = 0.364$; Y_3^3 constants); B-hodge-yubios.json (Eq. (33), Table 6); C-predictions.json (P-H1, P-H2, P-H3); D-construction-sensitivity.json (Table 7); E-harmonic-support.json (harmonic-energy concentration); E1-hodge-nulls-report.md (the Hodge report in full, including failures); F-hodge-cycle18-knn8.json ($N = 1391$ replication); ladder_VD.json (Table 1); finite_size.json, finite_size_colperm.json, scaling_VN.json (Table 4, Eq. (18)); signal_recovery.json (Table 8 and the false-positive block); gwtc.json (GWTC-like corpora and nulls); manifest.json (per-corpus N , column sums, zero-row counts, source paths); SUMMARY.md (ladder, recovery, N_c re-analysis, Y_3^3 constants).

Consolidated series (timeseries/new-series/).

INDEX.json (provenance, null convention, validation block); ladder_VD.json (77 rows, V_2 vs D); scaling_VN.json (31 rows, V_2 vs N , null-subtracted); signal_recovery.json (248 rows, planted-amplitude sweep plus FPR); gwtc-new.json (4 rows). Generated matrices are shipped in data/samples/ (family_i_latent_class.npz, family_ii_curved.npz, family_iii_gwtc.npz, MANIFEST.json); they are also regenerable from scripts/generate_samples.py in under a second.

Instrument (skills/curved-corpus-create/).

SKILL.md (conventions, gate failure mode, anti-patterns, red flags), scripts/create_corpus.py (subcommands generate, measure, calibrate, place; -selftest asserts seven blocks including the round trip of Eq. (40) and the K -invariance of Eq. (11)), references/ (worked calibration example). Dependencies: Python stdlib and numpy only, no network.

Real inputs.

The 2286×9 master matrix is shipped at data/real/per_row_coverage_v3.json (column sums $[1640, 1304, 1208, 1296, 1098, 1320, 1776, 1373, 1204]$, 204 zero rows); it originated as cycle-20/per_row_coverage_v3.json and earlier drafts of this appendix cited it under that historical path. The 1391×9 replication input (cycle-18/per_row_coverage.json, 14 zero rows) is *not* shipped in the bundle; only the results derived from it are, in results/F-hodge-cycle18-knn8.json. The headline V_2 is computed over all 2286 rows, the 204 all-zero rows included; restricting to the 2082 non-zero rows gives 0.678655 instead of 0.723529, so the inclusion policy is part of the definition of the number. Primitive order throughout: attestation, trust_chain, least_privilege, declarative_policy, continuous_adaptive, immutability, audit_evidence, cryptographic_identity, segmentation.

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